

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

5. Basic modeling and forecasting



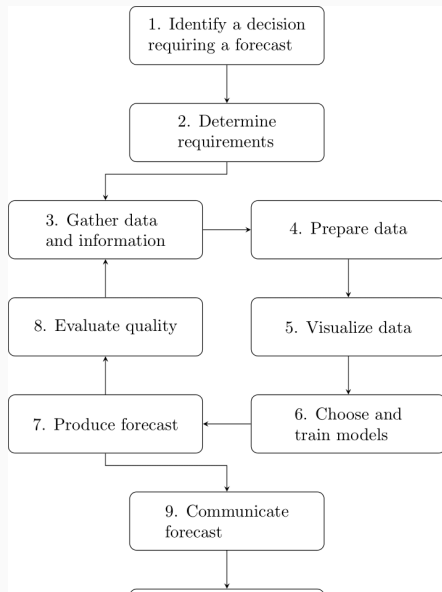
Outline

- 1 Statistical forecasting
- 2 What can we forecast?
- 3 Benchmark methods
- 4 Specify and estimate
- 5 Produce forecasts
- 6 Fitted values and residuals

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Forecasting workflow



Statistical forecasting steps

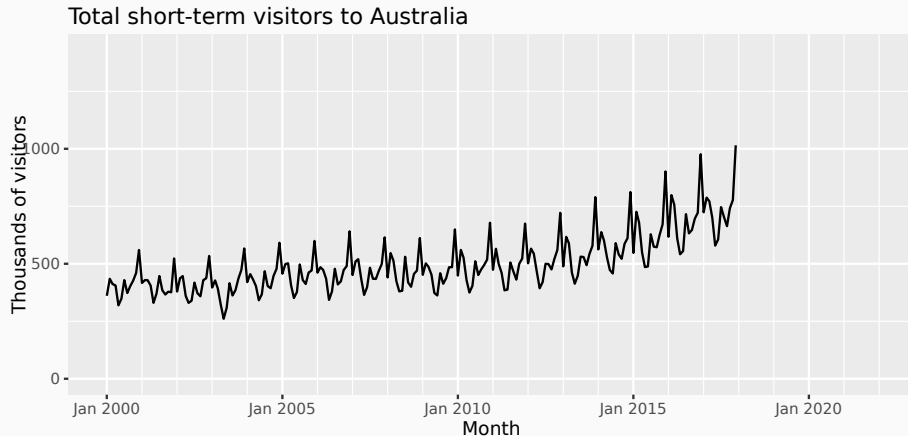
- Prepare data.
- Visualise data.
- Choosing and fitting models (specify and train models).
- Produce forecast.
- Evaluate quality.

What is a forecast?

A forecast is an estimate of the probability distribution of a variable to be observed in the future.

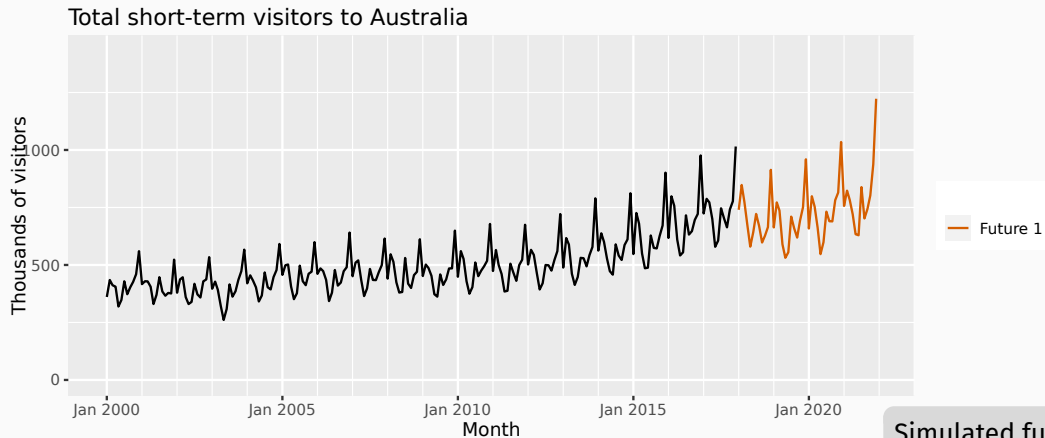
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What is a forecast?

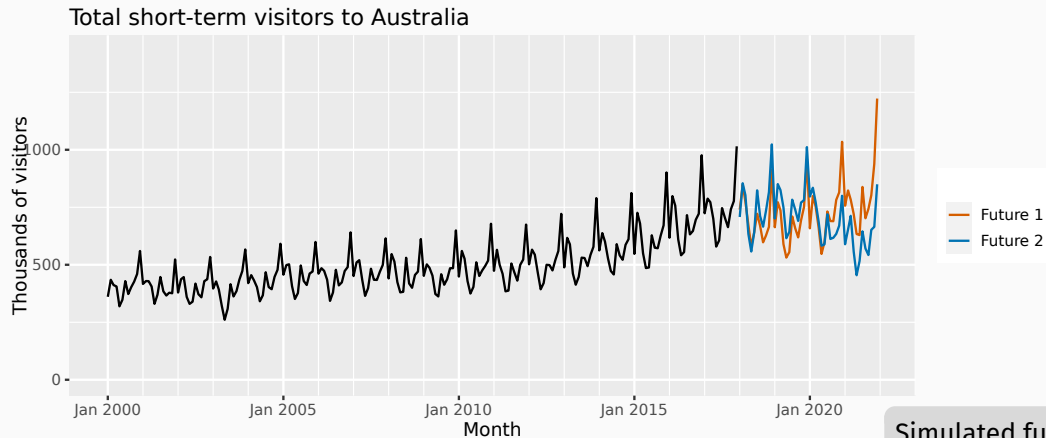
A forecast is an estimate of the probability distribution of a variable to be observed in the future.



Simulated futures
from an ETS model

What is a forecast?

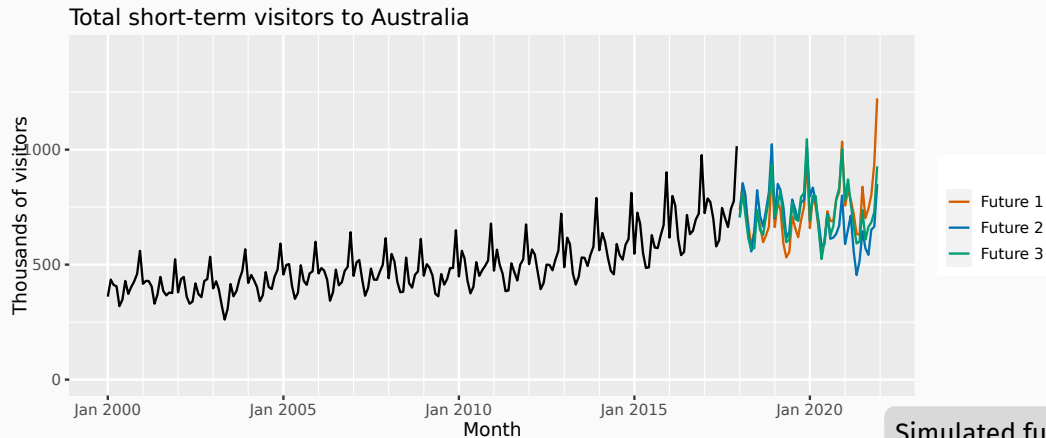
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Simulated futures
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What is a forecast?

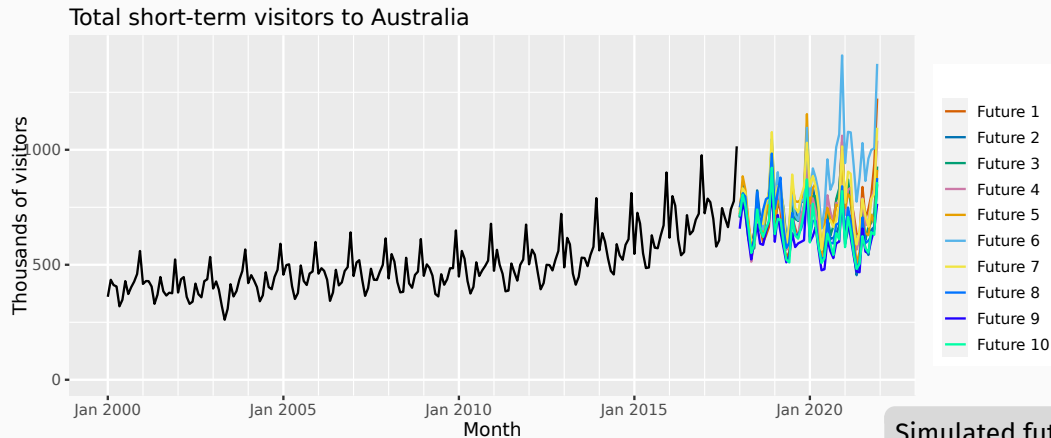
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Simulated futures
from an ETS model

What is a forecast?

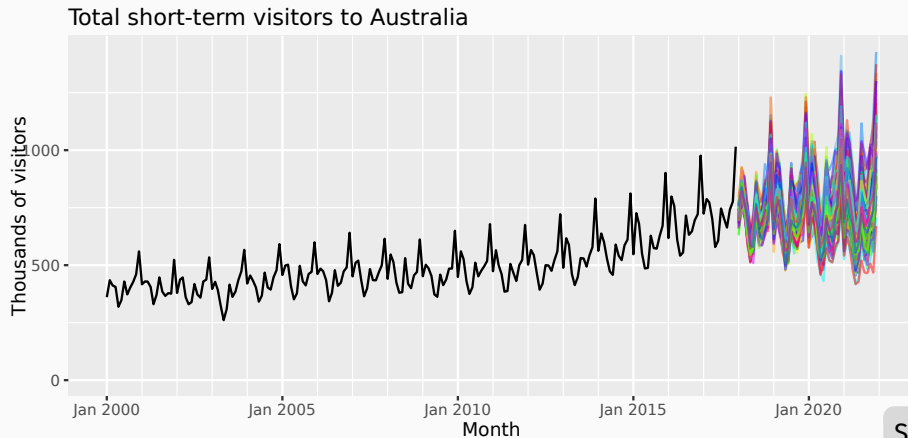
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Simulated futures
from an ETS model

What is a forecast?

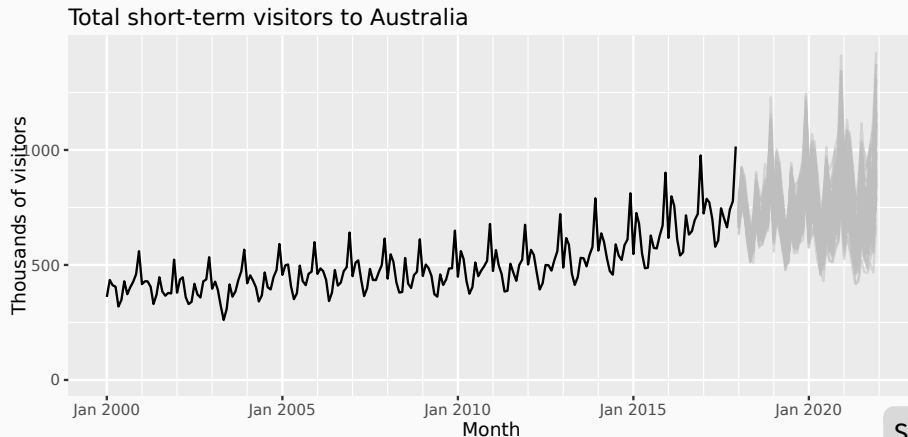
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Simulated futures
from an ETS model

What is a forecast?

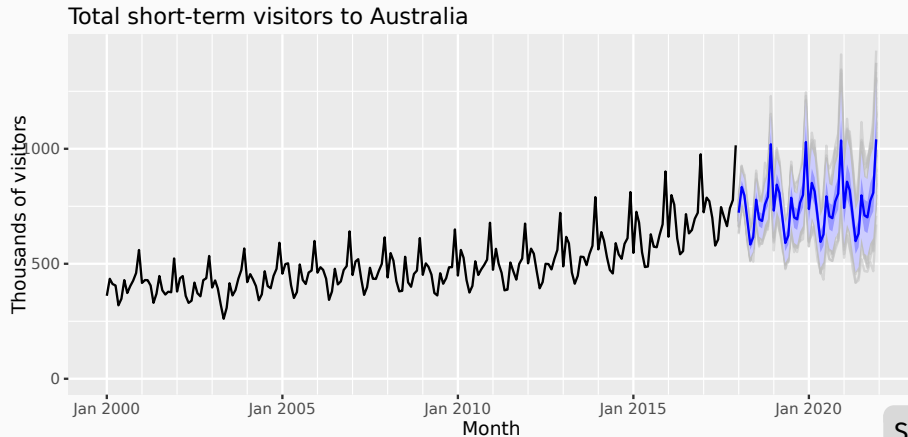
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Simulated futures
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What is a forecast?

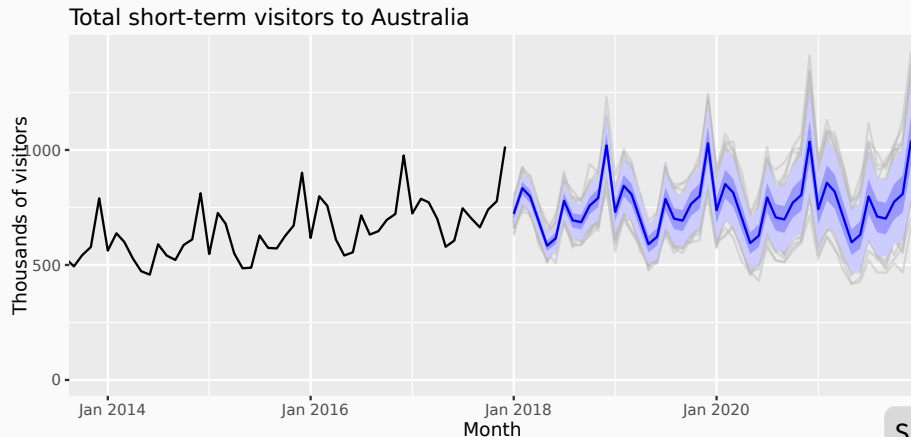
A forecast is an estimate of the probability distribution of a variable to be observed in the future.



Simulated futures
from an ETS model

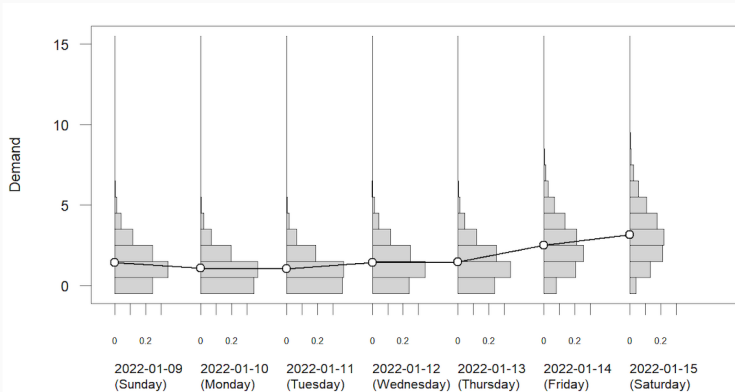
Prediction interval

A forecast is an estimate of the probability distribution of a variable to be observed in the future.

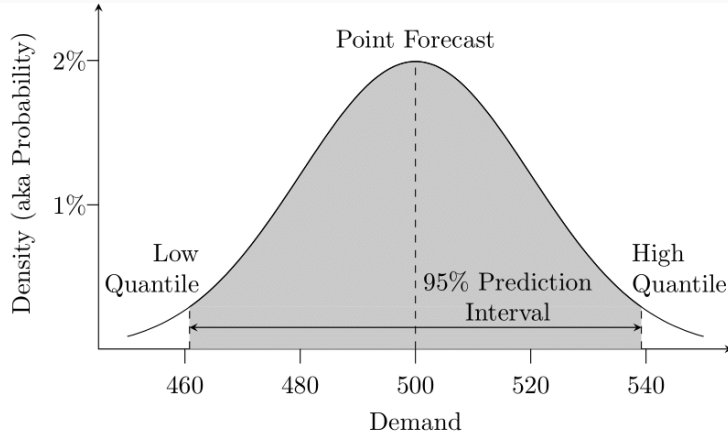


Simulated futures
from an ETS model

Visualising forecast distributions



Forecast distribution



Statistical forecasting

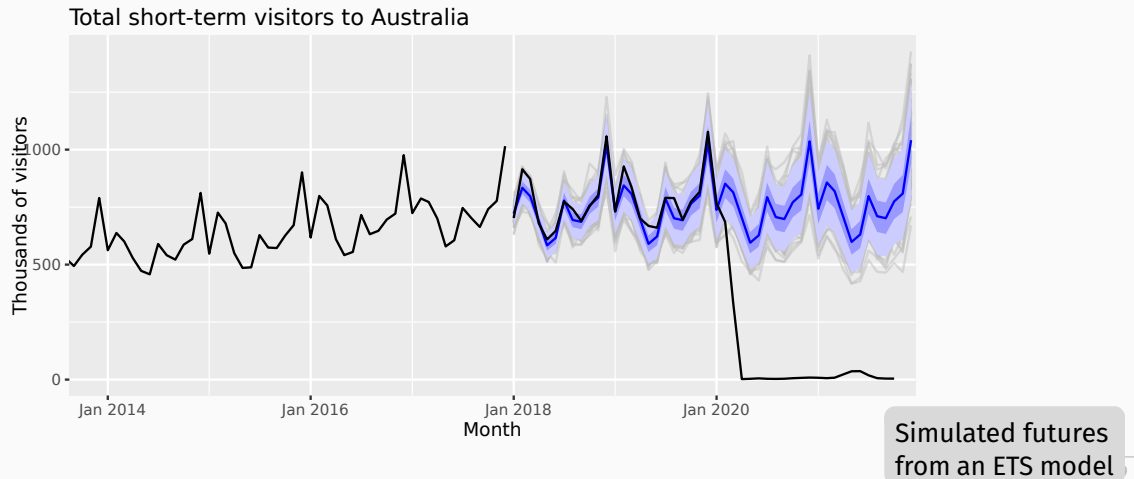
- Thing to be forecast: y_{T+h} .
- What we know: $y_1, \dots, y_T$.
- Forecast distribution: $y_{T+h|t} = y_{T+h} \mid \{y_1, y_2, \dots, y_T\}$.
- Point forecast: $\hat{y}_{T+h|T} = \mathbf{E}[y_{T+h} \mid y_1, \dots, y_T]$.
- Forecast variance: $\text{Var}[y_t \mid y_1, \dots, y_T]$
- Prediction interval is a range of values of y_{T+h} with high probability.

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What can we forecast?

A forecast is an estimate of the probability distribution of a variable to be observed in the future.



What can we forecast?



What can we forecast?



What can we forecast?



What can we forecast?



What can we forecast?

TOMORROW



What can we forecast?



What can we forecast?



Which is easiest to forecast?

- 1 daily electricity demand in 3 days time
- 2 timing of next Halley's comet appearance
- 3 time of sunrise this day next year
- 4 Google stock price tomorrow
- 5 Google stock price in 6 months time
- 6 maximum temperature tomorrow
- 7 exchange rate of \$US/AUS next week
- 8 total sales of drugs in Australian pharmacies next month

Which is easiest to forecast?

- 1 daily electricity demand in 3 days time
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- how do we measure “easiest”?
- what makes something easy/difficult to forecast?

Factors affecting forecastability

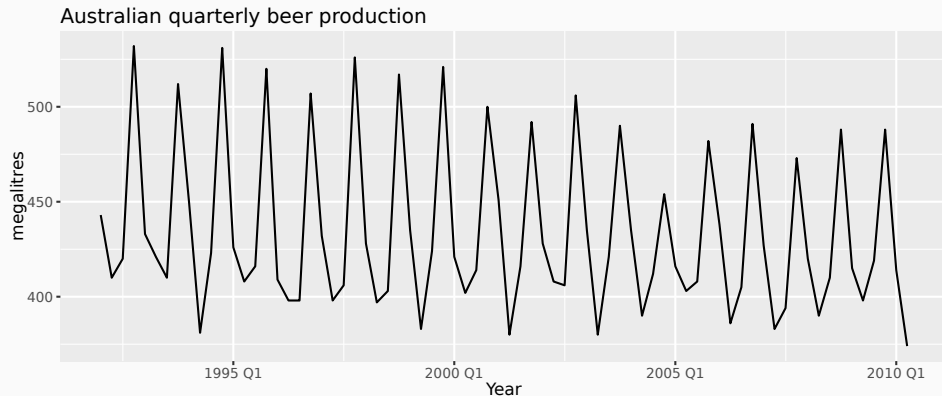
Something is easier to forecast if:

- we have a good understanding of the factors that contribute to it
- there is lots of data available;
- the forecasts cannot affect the thing we are trying to forecast.
- there is relatively low natural/unexplainable random variation.
- the future is somewhat similar to the past

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Some simple forecasting methods



How would you forecast these series?

Some simple forecasting methods



How would you forecast these series?

Some simple forecasting methods

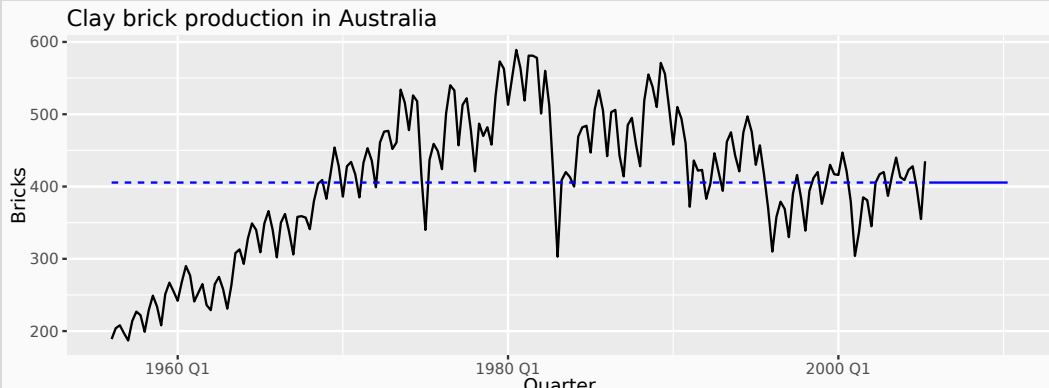


How would you forecast these series?

Some simple forecasting methods

MEAN(y): Average method

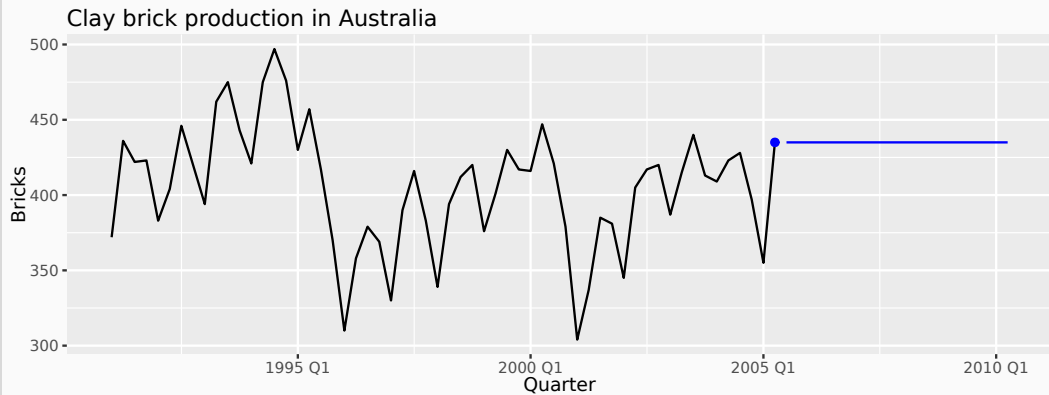
- Forecast of all future values is equal to mean of historical data $\{y_1, \dots, y_T\}$.
- Forecasts: $\hat{y}_{T+h|T} = \bar{y} = (y_1 + \dots + y_T)/T$



Some simple forecasting methods

NAIVE(y): Naïve method

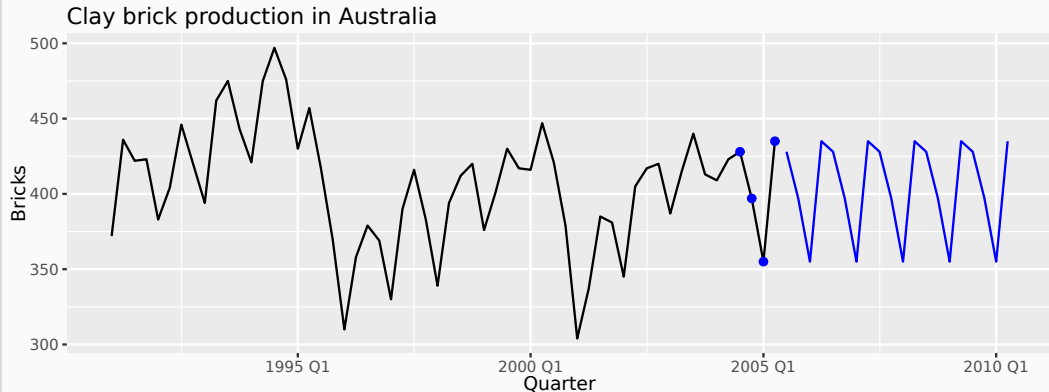
- Forecasts equal to last observed value.
- Forecasts: $\hat{y}_{T+h|T} = y_T$.
- Consequence of efficient market hypothesis.



Some simple forecasting methods

SNAIVE ($y \sim \text{lag}(m)$): Seasonal naïve method

- Forecasts equal to last value from same season.
- Forecasts: $\hat{y}_{T+h|T} = y_{T+h-m(k+1)}$, where m = seasonal period and k is the integer part of $(h-1)/m$.



Some simple forecasting methods

`RW(y ~ drift())`: Drift method

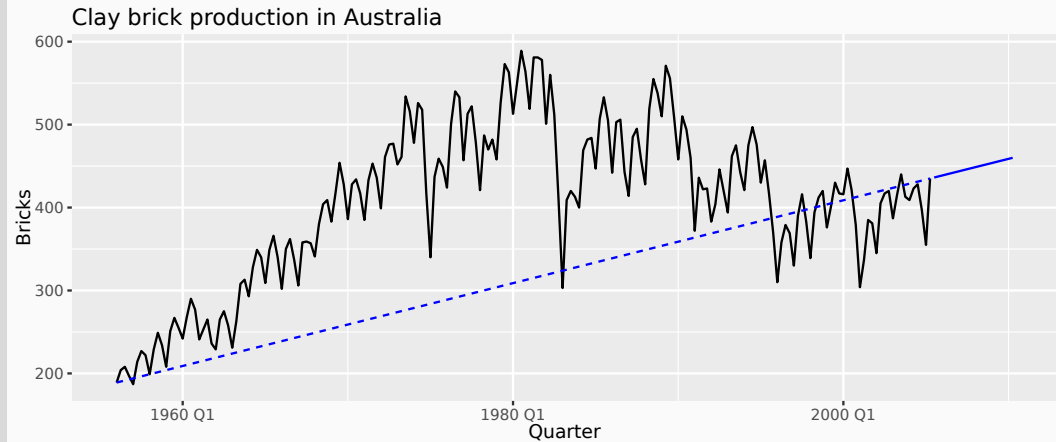
- Forecasts equal to last value plus average change.
- Forecasts:

$$\begin{aligned}\hat{y}_{T+h|T} &= y_T + \frac{h}{T-1} \sum_{t=2}^T (y_t - y_{t-1}) \\ &= y_T + \frac{h}{T-1} (y_T - y_1).\end{aligned}$$

- Equivalent to extrapolating a line drawn between first and last observations.

Some simple forecasting methods

Drift method



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Model specification

- Model specification in fable supports a formula based interface
- A model formula in R is expressed using `response ~ terms`
 - ▶ the formula's left side describes the response
 - ▶ the right describes terms used to model the response.
- Attention: **MODEL** name is in capital letters, e.g. `SNAIVE`

Model estimation

The `model()` function trains models on data. - It returns a mable object.

```
# Fit the models
my_mable <- my_data %>%
  model(
    choose_name1 = MODEL1(response ~ term1+...),
    choose_name2 = MODEL2(response ~ term1+...),
    choose_name3 = MODEL3(response ~ term1+...),
    choose_name4 = MODEL4(response ~ term1+...)
  )
```

Model fitting- example

The `model()` function trains models on data.

```
beer_fit <- aus_production |>
  model(
    `Seasonal_naïve` = SNAIVE(Beer),
    `Naïve` = NAIVE(Beer),
    Drift = RW(Beer ~ drift()),
    Mean = MEAN(Beer)
  )
```

```
# A mable: 1 x 4
  Seasonal_naïve   Naïve        Drift    Mean
      <model> <model>      <model> <model>
1      <SNAIVE> <NAIVE> <RW w/ drift> <MEAN>
```

A `mable` is a model table, each cell corresponds to a fitted model.

Extract information from mable

```
beer_fit %>% select(snaive) %>% report()  
beer_fit %>% tidy()  
beer_fit %>% glance()
```

- The `report()` function gives a formatted model-specific display.
- The `tidy()` function is used to extract the coefficients from the models.
- We can extract information about some specific model using the `filter()` and `select()` functions.

Check model performance

Once a model has been fitted, it is important to check how well it has performed on the data. I come back to this latter.

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Producing forecasts

- The `forecast()` function is used to produce forecasts from estimated models.
- **h** can be specified with:
 - ▶ a number (the number of future observations)
 - ▶ natural language (the length of time to predict)
 - ▶ provide a dataset of future time periods

Producing forecasts

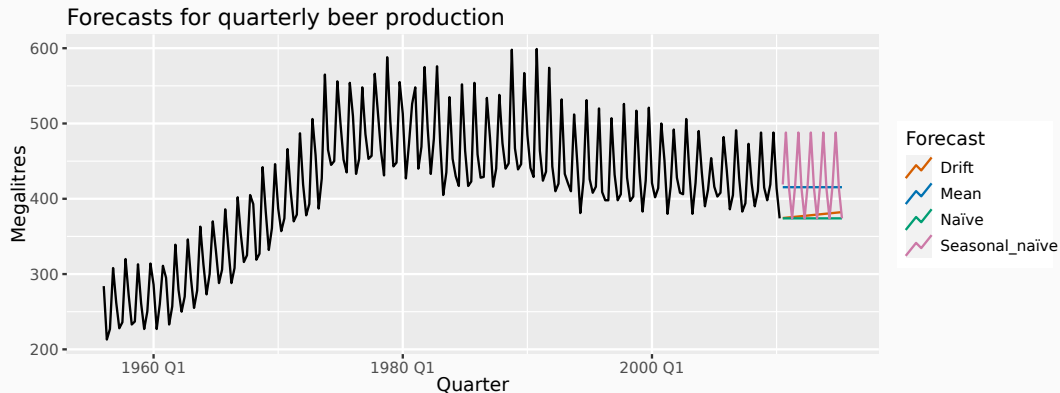
```
beer_fc <- beer_fit |>  
  forecast(h = "5 years")
```

```
# A fable: 80 x 4 [1Q]  
# Key:      .model [4]  
  .model      Quarter      Beer .mean  
  <chr>        <qtr>        <dist> <dbl>  
1 Seasonal_naïve 2010 Q3 N(419, 373) 419  
2 Seasonal_naïve 2010 Q4 N(488, 373) 488  
3 Seasonal_naïve 2011 Q1 N(414, 373) 414  
4 Seasonal_naïve 2011 Q2 N(374, 373) 374  
# i 76 more rows
```

A `fable` is a forecast table with point forecasts and distributions.

Visualising forecasts

```
beer_fc |>
  autoplot(aus_production, level = NULL) +
  labs(title = "Forecasts for quarterly beer production",
       x = "Quarter", y = "Megalitres") +
  guides(colour = guide_legend(title = "Forecast"))
```



Forecast distributions

- A forecast $\hat{y}_{T+h|T}$ is (usually) the mean of the conditional distribution $y_{T+h} \mid y_1, \dots, y_T$.
- Most time series models produce normally distributed forecasts.
- The forecast distribution describes the probability of observing any future value.

Forecast distributions - normal distribution

Assuming residuals are normal, uncorrelated, $\text{sd} = \hat{\sigma}$:

Mean: $\hat{y}_{T+h|T} \sim N(\bar{y}, (1 + 1/T)\hat{\sigma}^2)$

Naïve: $\hat{y}_{T+h|T} \sim N(y_T, h\hat{\sigma}^2)$

Seasonal naïve: $\hat{y}_{T+h|T} \sim N(y_{T+h-m(k+1)}, (k+1)\hat{\sigma}^2)$

Drift: $\hat{y}_{T+h|T} \sim N(y_T + \frac{h}{T-1}(y_T - y_1), h\frac{T+h}{T}\hat{\sigma}^2)$

where k is the integer part of $(h-1)/m$.

Note that when $h = 1$ and T is large, these all give the same approximate forecast variance: $\hat{\sigma}^2$.

Forecast distributions from bootstrapping

When a normal distribution for the residuals is an unreasonable assumption, one alternative is to use bootstrapping, which only assumes that the residuals are uncorrelated with constant variance.

- A one-step forecast error is defined as $e_t = y_t - \hat{y}_{t|t-1}$,
 $y_t = \hat{y}_{t|t-1} + e_t$.
- So we can simulate the next observation of a time series using $y_{T+1} = \hat{y}_{T+1|T} + e_{T+1}$
- Adding the new simulated observation to our data set, we can repeat the process to obtain $y_{T+2} = \hat{y}_{T+2|T+1} + e_{T+2}$

Generate many possible future using generate()

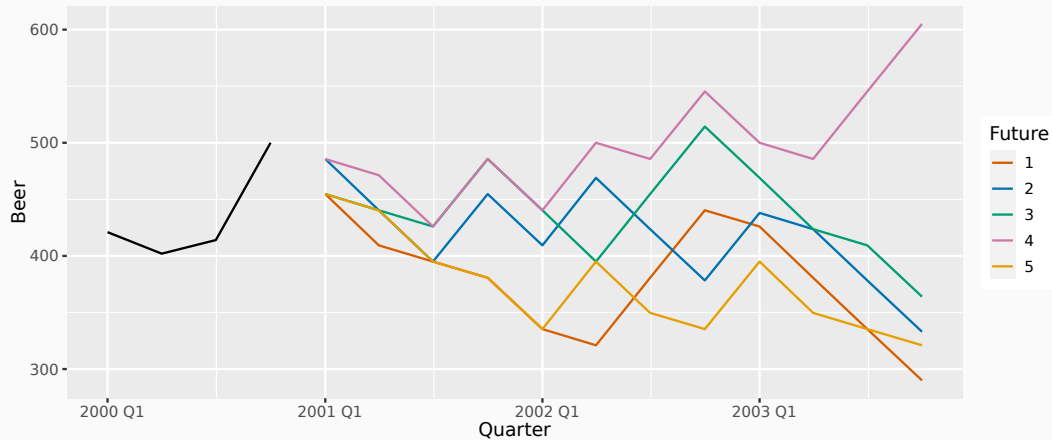
```
beer_2000 <- aus_production |> filter(year(Quarter) == 2000) |> select(Beer)
fit <- beer_2000 |>
  model(NAIVE(Beer))
sim <- fit |> generate(h = 12, times = 5, bootstrap = TRUE)
sim
```

```
# A tsibble: 60 x 5 [1Q]
```

```
# Key:       .model, .rep [5]
```

	.model	.rep	Quarter	.innov	.sim
	<chr>	<chr>	<qtr>	<dbl>	<dbl>
1	NAIVE(Beer)	1	2001 Q1	-45.3	455.
2	NAIVE(Beer)	1	2001 Q2	-45.3	409.
3	NAIVE(Beer)	1	2001 Q3	-14.3	395
4	NAIVE(Beer)	1	2001 Q4	-14.3	381.
5	NAIVE(Beer)	1	2002 Q1	-45.3	335.
6	NAIVE(Beer)	1	2002 Q2	-14.3	321

Generate 5 different futures



casts

```
fc <- fit |> forecast(h = 12, bootstrap = TRUE)
fc
```

```
# A fable: 12 x 4 [1Q]
```

```
# Key:      .model [1]
```

	.model	Quarter	Beer	.mean
	<chr>	<qtr>	<dist>	<dbl>
1	NAIVE(Beer)	2001 Q1	sample[5000]	500.
2	NAIVE(Beer)	2001 Q2	sample[5000]	500.
3	NAIVE(Beer)	2001 Q3	sample[5000]	500.
4	NAIVE(Beer)	2001 Q4	sample[5000]	499.
5	NAIVE(Beer)	2002 Q1	sample[5000]	500.
6	NAIVE(Beer)	2002 Q2	sample[5000]	500.
7	NAIVE(Beer)	2002 Q3	sample[5000]	500.
8	NAIVE(Beer)	2002 Q4	sample[5000]	500.

Prediction intervals

- Forecast intervals can be extracted using the `hilo()` function.

```
fit <- aus_production |> select(Beer) %>% model(NAIVE(Beer))  
forecast(fit) %>% hilo(level = 95) %>% unpack_hilo("95%")
```

```
# A tsibble: 8 x 6 [1Q]
```

```
# Key:       .model [1]
```

	.model	Quarter	Beer	.mean	`95%_lower`	`95%_upper`
	<chr>	<qtr>	<dist>	<dbl>	<dbl>	<dbl>
1	NAIVE(Beer)	2010 Q3	N(374, 4580)	374	241.	507.
2	NAIVE(Beer)	2010 Q4	N(374, 9159)	374	186.	562.
3	NAIVE(Beer)	2011 Q1	N(374, 13739)	374	144.	604.
4	NAIVE(Beer)	2011 Q2	N(374, 18319)	374	109.	639.
5	NAIVE(Beer)	2011 Q3	N(374, 22898)	374	77.4	671.

Prediction intervals

```
beer_fc |>  
  hilo(level = c(50, 75))
```

```
# A tsibble: 80 x 6 [1Q]
```

```
# Key:           .model [4]
```

	.model	Quarter	Beer	.mean	`50%`	`75%`
	<chr>	<qtr>	<dist>	<dbl>	<hilo>	<hilo>
1	Seasonal_naïve	2010 Q3	N(419, 373)	419	[406, 432]	50 [397, 441] 75
2	Seasonal_naïve	2010 Q4	N(488, 373)	488	[475, 501]	50 [466, 510] 75
3	Seasonal_naïve	2011 Q1	N(414, 373)	414	[401, 427]	50 [392, 436] 75
4	Seasonal_naïve	2011 Q2	N(374, 373)	374	[361, 387]	50 [352, 396] 75
5	Seasonal_naïve	2011 Q3	N(419, 747)	419	[401, 437]	50 [388, 450] 75
6	Seasonal_naïve	2011 Q4	N(488, 747)	488	[470, 506]	50 [457, 519] 75
7	Seasonal_naïve	2012 Q1	N(414, 747)	414	[396, 432]	50 [383, 445] 75
8	Seasonal_naïve	2012 Q2	N(374, 747)	374	[356, 392]	50 [343, 405] 75
9	Seasonal_naïve	2012 Q3	N(419, 1120)	419	[396, 442]	50 [380, 458] 75

Prediction intervals

```
beer_fc |>  
  hilo(level = c(50, 75)) |>  
  mutate(lower = `50%`$lower, upper = `50%`$upper)
```

```
# A tsibble: 80 x 8 [1Q]
```

```
# Key:           .model [4]
```

	.model <chr>	Quarter <qtr>	Beer <dist>	.mean <dbl>	`50%` <hilo>	`75%` <hilo>	lower <dbl>	upper <dbl>
1	Seasonal_naïve	2010 Q3	N(419, 373)	419	[406, 432]	50 [397, 441]	75 406.	432.
2	Seasonal_naïve	2010 Q4	N(488, 373)	488	[475, 501]	50 [466, 510]	75 475.	501.
3	Seasonal_naïve	2011 Q1	N(414, 373)	414	[401, 427]	50 [392, 436]	75 401.	427.
4	Seasonal_naïve	2011 Q2	N(374, 373)	374	[361, 387]	50 [352, 396]	75 361.	387.
5	Seasonal_naïve	2011 Q3	N(419, 747)	419	[401, 437]	50 [388, 450]	75 401.	437.
6	Seasonal_naïve	2011 Q4	N(488, 747)	488	[470, 506]	50 [457, 519]	75 470.	506.
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8	Seasonal_naïve	2012 Q2	N(374, 747)	374	[356, 392]	50 [343, 405]	75 356.	392.

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Fitted values

- $\hat{y}_{t|t-1}$ is the forecast of y_t based on observations $y_1, \dots, y_t$.
- We call these “fitted values”.
- Sometimes drop the subscript: $\hat{y}_t \equiv \hat{y}_{t|t-1}$.
- Often not true forecasts since parameters are estimated on all data.

For example:

- $\hat{y}_t = \bar{y}$ for average method.
- $\hat{y}_t = y_{t-1} + (y_T - y_1)/(T - 1)$ for drift method.

Forecasting residuals

Residuals in forecasting: difference between observed value and its fitted value: $e_t = y_t - \hat{y}_{t|t-1}$.

Beer production - augment

```
fit <- aus_production |> select(Beer) %>% model(SNAIVE(Beer))  
augment(fit)
```

```
# A tsibble: 218 x 6 [1Q]
```

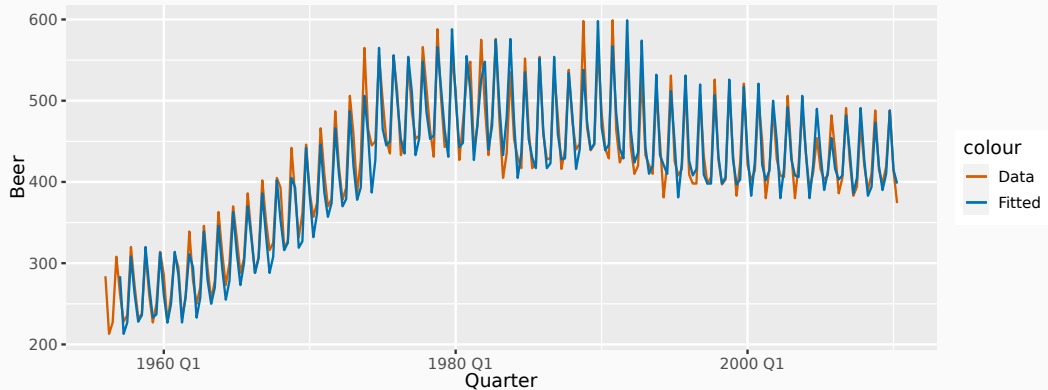
```
# Key:           .model [1]
```

	.model <chr>	Quarter <qtr>	Beer <dbl>	.fitted <dbl>	.resid <dbl>	.innov <dbl>
1	SNAIVE(Beer)	1956 Q1	284	NA	NA	NA
2	SNAIVE(Beer)	1956 Q2	213	NA	NA	NA
3	SNAIVE(Beer)	1956 Q3	227	NA	NA	NA
4	SNAIVE(Beer)	1956 Q4	308	NA	NA	NA
5	SNAIVE(Beer)	1957 Q1	262	284	-22	-22
6	SNAIVE(Beer)	1957 Q2	228	213	15	15
7	SNAIVE(Beer)	1957 Q3	236	227	9	9
8	SNAIVE(Beer)	1957 Q4	320	308	12	12
9	SNAIVE(Beer)	1958 Q1	272	262	10	10
10	SNAIVE(Beer)	1958 Q2	233	228	5	5

```
# i 208 more rows
```

Beer production - fitted values

```
augment(fit) |>  
  ggplot(aes(x = Quarter)) +  
  geom_line(aes(y = Beer, colour = "Data")) +  
  geom_line(aes(y = .fitted, colour = "Fitted"))
```



Beer production - residuals

```
augment(fit) |>  
  autoplot(.resid) +  
  labs(x = "Quarter", y = "", title = "Residuals from snaïve method")
```

